

**T Level Technical Qualification in Digital Software Development (Level 3)****Additional Specimen Assessment Material****Time** 3 hours**Paper
reference****XXXXXX/XX****Core: Employer Set Project****TASK 2: Identifying and fixing defects in existing code
Task Booklet****You must have:***Task2_RetailX_NonWorkingCode.py**Task2_Test_Log_Template.docx***Information**

- The total mark for this set task is 21.
- This booklet is specific to each series and this material must only be issued to students who have been entered to undertake the task in the relevant series.
- This booklet must be kept securely until the start of the timetabled assessment.

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Instructions to Students

You must complete ALL parts of the activity within the assessment.

Read the assessment information carefully.

You must plan your time and submit all required evidence at the end of the supervised period.

The task must be undertaken at the time and on the date specified by Pearson.

You have **three hours** in total to complete this task.

Your provider will advise you of when any supervised breaks have been scheduled.

The task must be completed under supervised conditions.

You must work independently throughout the supervised assessment period and should not share your work with other students.

During this task you are **not** permitted:

- to have access to the internet
- to use AI or any other tool designed to automate the testing and/or fixing process.

You are permitted to use **offline** versions of relevant software to produce evidence for this task.

Files provided for use during this activity:

- *Task2_Test_Log_Template.docx*
- *Task2_RetailX_NonWorkingCode.py*

Your work and any material provided must be kept securely at all times.

Set Task Brief

You are a member of the programming team that is developing a program to meet the requirements in the Set Task Information.

Your manager has asked you to look at some code that a Junior Software Developer has produced but which is not yet functional.

The code that is not yet functioning is provided for you in the file *Task2_RetailX_NonWorkingCode.py*.

Activity

Produce and apply a test plan to:

- identify the defects that exist in the code
- apply a solution to fix the defects in the code provided
- document the process that you followed to fix the code.

When applying a solution to fix the defects you must:

- ensure the code meets the requirements in the Set Task Information
- use Python 3 (*Python 3.10 upwards*) programming language
- follow accepted programming conventions
- test your solution to ensure that it functions as expected.

Use the Set Task Information, test log template and code provided when completing this activity.

(Total for Task 2 = 21 marks)

Outcomes for Submission

Save your code as a .py file.

Save your testing document as a PDF file.

All files should be saved in your folder for submission.

Use this naming convention:

- *Task2_DSD_Code_[Registration number]_[surname]_[first letter of first name]*
- *Task2_DSD_Test_Log_[Registration number]_[surname]_[first letter of first name]*

Set Task Information

RetailX Ltd has provided you with this set of information to develop your program.

Requirements

The program will be part of the new system's *Customer management and rewards module*.

You need to create a program that will allow the user to enter this information:

- customer's loyalty card number
- the value (in GBP) of each item in the transaction
- the product category of each item
- if the transaction is in-store or online.

The program should then provide the user with a summary of the transaction that shows:

- the customer's loyalty card number
- the total value of the transaction
- the loyalty points earned on each product
- subtotals showing:
 - the points earned before bonus points are added
 - the number of bonus points earned
- the overall total points earned.

Values that represent currency should be formatted as GBP to two decimal places.

Points earned are calculated using whole pounds (£) only.

The program should be able to handle user errors.

Loyalty card number

Must be eight numeric characters only.

Loyalty points reward structure

Product category	Information
Home electrical	Points earned per item: • 4 points per pound up to the value of £500 • 2 points per pound for any remaining value over £500
Computing and gaming	Points earned per item: • 4 points per pound
Accessories and consumables	Points earned per item: • 3 points per pound
An additional 2 points per pound on the total value of items is earned for in-store transactions.	

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